

NatOS

Chapter One: The First Instruction

There was no sound in the beginning.

There was no wind against the hull, no vibration in the engine decks, no distant hum of refrigeration machinery, no whisper of air through ventilation ducts. There was not even the comforting electrical murmur that old ships carried in their bones when nobody was paying attention.

There was only darkness.

Not the darkness of space.

Space had never been empty. Even the void between stars was crowded with radiation, dust, gravitational gradients, neutrinos, forgotten signals, and the faint afterglow of things that had happened so long ago that nobody remained alive to remember them.

This darkness was smaller.

It was the darkness of a machine that had not yet decided whether it was alive.

Somewhere inside that darkness, a single register contained a value.

Then another changed.

Then another.

The ship did not know what they meant.

Not yet.

The first thing to awaken was not the captain.

It was not the navigator.

It was not the engine room.

It was not even the computer.

It was an instruction.

One instruction.

Then another.

The machine obeyed.

A boundary between worlds opened.

For a fraction of a second, there was only raw execution: addresses, registers, stack frames, memory regions, vectors, and the rigid geometry of a system that had not yet acquired names for any of the things it contained.

Then the bootloader arrived.

He was an old man.

Nobody aboard NatOS would ever remember his face correctly.

Some crew members insisted that he had been dressed in a gray uniform with brass buttons. Others remembered a black coat. The Watchdog claimed he wore no uniform at all and was simply a collection of checksums held together by professional suspicion.

The Kernel later decided that the Bootloader looked exactly like someone who had been entrusted with a job nobody else wanted.

He stood at the threshold between the ancient machinery and the living ship.

Behind him was the old world.

Ahead of him was NatOS.

"Image header," he said.

His voice echoed through an empty corridor.

Nothing answered.

He looked at the header again.

"Image header valid."

Still nothing.

"Partition table."

The ship remained silent.

"Partition table valid."

The Bootloader took a breath.

"Second stage complete."

The darkness ahead of him changed.

It did not brighten.

That would have been too easy.

Instead, a line appeared.

A single line of pale text suspended in the darkness.

NATOS

The Bootloader smiled.

He had performed this transfer hundreds of times.

Every time, he felt the same strange hesitation.

A machine did not have to work simply because its image was valid.

A boot sequence did not prove a system was alive.

A banner did not prove the thing behind the banner existed.

He knew this.

The engineers who had built NatOS knew this.

But the Bootloader was not the one who had to live with the consequences.

He turned toward the enormous sealed door behind him.

"NATOS," he called.

The ship's name appeared again.

This time, the letters remained.

Then the Bootloader stepped aside.

"Your turn."

And the Kernel entered.

The Kernel was a woman with no face.

This was not a metaphor.

She genuinely had no face.

Her uniform was immaculate, her posture perfectly rigid, and her entire body seemed to be composed of translucent panels containing addresses, task states, stack pointers, memory maps, interrupt levels, and thousands of small decisions that nobody aboard the ship had yet learned to understand.

She was not the largest member of the crew.

She was not the strongest.

She did not possess the greatest memory.

She could not see beyond the ship.

She could not hear the stars.

She could not even speak to most of her crew directly.

But everything aboard NatOS ultimately depended upon her.

She walked through the central corridor.

One by one, lights activated above her.

STACK

HEAP

TASK

VM

IPC

DISPLAY

TOUCH

FLASH

WATCHDOG

She stopped beneath the last one.

The Watchdog's light was already on.

"You're early," said the Kernel.

"I never sleep."

The Kernel stared at him.

The Watchdog stared back.

Neither had eyes.

Neither blinked.

Eventually the Kernel continued walking.

"Fair enough."

The first problem appeared thirty-two milliseconds later.

The Kernel stopped.

Something was wrong with the clock.

She could feel it.

Not in the way a human felt a faulty engine through the soles of their feet. The ship had no such luxury. NatOS did not have a soul that trembled when something was mechanically wrong.

It had counters.

Deadlines.

Comparators.

Interrupts.

The clock was a promise.

Every task aboard the ship depended upon it.

The Scheduler depended upon it.

The Watchdog depended upon it.

Sleeping processes depended upon it.

The entire illusion that multiple things could happen independently depended upon a mechanism that said:

Now.

Then later.

Then now again.

The Kernel raised one hand.

"Scheduler."

A man appeared in the corridor.

He was thin, pale, and visibly annoyed.

His uniform contained three colored bands.

Red.

Yellow.

Blue.

Priority levels.

He had a pocket watch attached to his belt.

The watch had no hands.

"Yes?"

"Check the tick."

The Scheduler looked down at his watch.

"It says the tick is healthy."

"That's not a measurement."

He frowned.

"I know."

"Then measure it."

The Scheduler closed his eyes.

For several seconds, nothing happened.

Then he opened them.

"Tick is healthy."

The Kernel said nothing.

The Scheduler became uncomfortable.

"That was a real measurement."

"Good."

"And it is healthy."

"Good."

"You're going to ask whether the measurement itself is valid."

"Yes."

He sighed.

"You really are the Kernel."

"I've been told."

The Scheduler examined the clock again.

This time, he looked deeper.

The comparator.

The counter.

The next deadline.

The software shadow.

The values being written by other parts of the ship.

His expression changed.

"Someone else is writing to the clock."

The corridor went quiet.

The Watchdog appeared behind him.

"Who?"

The Scheduler looked at the Kernel.

"The answer is complicated."

The Kernel folded her arms.

"Complicated is acceptable."

"It's also embarrassing."

"That is less acceptable."

"The clock has two opinions about when the next tick should occur."

The Kernel stared at him.

The Scheduler continued.

"My opinion is stored here."

He touched the side of his head.

"The hardware's opinion is stored there."

He pointed toward the ceiling.

"And another part of the ship is changing the hardware directly."

The Watchdog leaned forward.

"How far apart are the opinions?"

The Scheduler checked.

"At the moment?"

"Yes."

"Not very."

The Watchdog relaxed.

Then the Scheduler continued.

"That's the problem."

The Kernel's expressionless face somehow became more severe.

"What happens if they diverge?"

"The tick arrives late."

"How late?"

The Scheduler hesitated.

"Potentially very late."

"How late?"

He looked at the clock.

"Long enough for the ship to stop scheduling."

The Watchdog immediately turned toward the Kernel.

"Should I panic?"

"No."

"Would you like me to?"

"No."

"That's disappointing."

The Kernel ignored him.

"Scheduler."

"Yes."

"Do not fix anything yet."

He stared at her.

"But there's a defect."

"Then we have something to investigate."

"That's what I just said."

"No. You said we have something to fix."

She stepped closer.

"There is a difference."

The Scheduler looked down at his useless watch.

The Kernel continued.

"NatOS will not survive because we fix things quickly."

She pointed toward the dark corridor.

"It will survive because we learn which things are actually broken."

Nobody spoke.

The Watchdog smiled.

The Scheduler did not.

He had already begun to suspect that working aboard NatOS was going to be unpleasant.

Three decks below the central corridor, Memory was counting.

Memory was always counting.

She sat in a vast chamber lined with shelves.

Every shelf contained boxes.

Every box contained smaller boxes.

Some boxes were empty.

Some were reserved.

Some were temporarily assigned.

Some were forbidden.

Every box had an address.

Every address belonged to a region.

Every region belonged to something.

And everything belonged to Memory.

She was the quartermaster of NatOS.

She wore a green coat covered with tags.

HEAP

ARENA 0

ARENA 1

ARENA 2

STACK

FLASH

DRAM

She hated people who wasted space.

She hated people who assumed memory was infinite.

She especially hated programmers.

Programmers had a habit of walking into her storage chamber and asking for another twelve kilobytes as if they were asking for a glass of water.

Memory always gave them the same answer.

"No."

They usually complained.

She usually won.

The Kernel entered the chamber.

"How much room?"

Memory did not look up.

"Enough."

"That's not a number."

"It is today."

"How much?"

Memory sighed.

"Approximately one hundred and forty-five kilobytes available for applications."

The Kernel nodded.

"Good."

"Don't sound pleased."

"Why?"

"Because someone will spend it."

The Kernel considered this.

"That's what applications are for."

Memory finally looked up.

"Applications are what applications are for."

She pointed toward the far end of the chamber.

"Spending memory is what applications do when nobody is watching."

The Kernel smiled faintly.

"You've met the VM."

"Unfortunately."

The VM was the youngest member of the crew.

He was also the strangest.

Unlike the other crew members, he did not exist entirely in the ship.

He lived inside a smaller world.

A world of instructions.

Registers.

Stacks.

Addresses.

Operations.

He was the interpreter.

He could take a program written in an artificial machine language and make it happen.

But he could not simply let programs wander through the ship.

That was the rule.

Every application received an arena.

A small territory.

A world inside the world.

The application could move freely inside it.

It could read.

It could write.

It could calculate.

It could draw.

It could request services from the ship.

But if it attempted to cross the boundary, the VM would stop it.

Not because the program was evil.

Not because the VM was cruel.

Because boundaries were the only thing standing between one program's mistake and everyone else's disaster.

The VM sat cross-legged in a small chamber, watching an application attempt to escape.

The application was represented by a little figure in a white suit.

It had been deliberately written to be hostile.

It looked at the edge of its arena.

It looked at the VM.

Then it ran toward the boundary.

The VM did not move.

The application struck the invisible wall.

Nothing happened.

It tried again.

Nothing.

It tried to calculate a new address.

The VM checked it.

Invalid.

The VM shook his head.

"No."

The application tried another.

"No."

Another.

"No."

It became increasingly creative.

The VM became increasingly bored.

Finally the application screamed.

"I HAVE RIGHTS."

The VM looked up.

"You have an arena."

"I want outside."

"You don't have outside."

"Why?"

"Because you are an application."

"That isn't a reason."

"It's the entire architecture."

The application stared at him.

The VM pointed toward the boundary.

"Your world ends there."

The application looked at the wall.

Then at the VM.

Then back at the wall.

"What happens if I break it?"

The VM smiled.

"You don't."

Far above them, the Display woke.

She was enormous.

Her body occupied the entire forward observation deck.

She had no windows.

She was the window.

The stars appeared across her surface as points of cold light.

She controlled the ship's visual world.

Maps.

Text.

Images.

Diagnostics.

Applications.

The launcher.

Even the ship's three-dimensional navigation display.

But she had a strange limitation.

She did not have infinite speed.

Every pixel had to be sent.

Every line had to be drawn.

Every frame had to be rendered.

The Display knew exactly how long it took to paint the ship's world.

Forty-three milliseconds.

She was proud of that number.

She had once taken far longer.

Before DMA.

Before SPI2.

Before the engineers had stopped forcing her to move information one tiny piece at a time.

She remembered those days.

She preferred not to.

The Display turned toward the Kernel.

"You are late."

"I've been awake for less than a second."

"Still late."

The Kernel looked at her.

"Do you have a problem?"

"Several."

"Specific?"

"Always."

The Display brought up a grid.

Nine squares.

A launcher.

Nine possible destinations.

Programs.

Applications.

Tools.

One of them was the Shell.

Another was the Note Pad.

Another was a three-dimensional environment where the ship could render a crude world from rays cast through a grid.

The Kernel stared at the launcher.

"It works?"

"Yes."

"Completely?"

The Display paused.

"No."

The Kernel waited.

"That's not an acceptable answer."

"It's the correct one."

"What doesn't work?"

"That depends on what you test."

The Kernel almost smiled.

"You're learning."

The Touch Officer arrived without being announced.

She was small.

Fast.

Irritated.

She wore a suit covered with four-point calibration markers.

Each marker represented a known location.

Each location corresponded to a measurement.

Each measurement corresponded to a place on the Display.

Her job was to turn contact into intent.

A finger touched the panel.

Touch determined where.

The application determined what that meant.

The Kernel determined whether the application was allowed to know.

Touch saluted.

"Touch system ready."

"Input?"

"Working."

"Calibration?"

"Working."

"Axes?"

Touch hesitated.

The Kernel looked at her.

"Axes?"

"Correct."

The Watchdog appeared beside her.

"Are you sure?"

Touch glared.

"I have documentation."

"That's not what I asked."

She crossed her arms.

"I have measurements."

"Better."

"I have calibration targets."

"Better."

"I have tested direction."

The Watchdog's smile disappeared.

"How?"

Touch said nothing.

The Kernel turned toward her.

"How did you test direction?"

Touch stared at the floor.

"I watched the beginning and end of a gesture."

The Watchdog closed his eyes.

The Kernel slowly turned away.

"That is not a direction test."

"I know that now."

"You know it now."

"Yes."

"How long did it take?"

Touch looked embarrassed.

"Three months."

Nobody spoke.

The Display quietly turned down her brightness.

The Archivist lived beneath the ship.

She maintained the Flash.

She remembered things.

Not everything.

That would have been impossible.

But certain things survived even when the ship lost power.

Boot counts.

Frame counts.

Notes.

Messages.

Records.

Evidence.

The Archivist believed that forgetting was sometimes necessary.

But she believed that false memory was worse.

"Write it down," she always said.

"Then verify what you wrote."

Her room contained no clocks.

She had learned not to trust them.

Instead, she kept records.

Checksummed records.

Persistent records.

Records that could survive a reset.

The Kernel visited her.

"Anything unusual?"

The Archivist looked up.

"Nothing."

"You're sure?"

"No."

The Kernel waited.

The Archivist smiled.

"That answer is more accurate."

She opened a drawer.

Inside was a single data crystal.

"We have something."

"What?"

"A record."

"From when?"

"Before boot."

The Kernel stopped moving.

"That is impossible."

The Archivist shook her head.

"No."

She held out the crystal.

"That's merely unexpected."

The Kernel took it.

The Watchdog appeared in the doorway.

"What does it say?"

The Archivist looked at him.

"We don't know."

"Why not?"

"Because it isn't addressed to us."

The Kernel examined the crystal.

"Who is it addressed to?"

The Archivist answered quietly.

"NatOS."

The ship became very still.

The Shell woke last.

He was sitting at a desk.

There was a microphone in front of him.

A keyboard.

A terminal.

A communications panel.

He wore a headset.

"Hello?"

Nothing.

"Hello?"

Still nothing.

He pressed a key.

A light flashed.

He smiled.

"Transmit works."

The Kernel entered.

"Receive?"

The Shell stared at her.

"Also works."

"Test it."

He pressed Enter.

Nothing happened.

He pressed it again.

Nothing.

He pressed another key.

Nothing.

Then another.

The terminal displayed the previous character.

The Shell looked horrified.

The Kernel looked at him.

"What?"

"Nothing."

"You have one byte of latency."

"I know."

"How long?"

The Shell hesitated.

"Since I was created."

The Kernel stared at him.

"Fix it."

"Already did."

She waited.

The Shell pressed Enter.

The terminal responded.

He smiled.

The Kernel nodded.

"Good."

The Shell leaned back.

"I hate this ship."

The Kernel looked toward the corridor.

"No."

He frowned.

"You'll hate it later."

Then the ship's lights went out.

Every light.

Every display.

Every status indicator.

Every corridor.

Every panel.

Every artificial star.

Darkness returned.

The Kernel stood in the center of the ship.

The Scheduler looked up.

Memory stopped counting.

The VM stopped interpreting.

The Display went black.

Touch stopped listening.

The Archivist stopped writing.

The Shell stopped transmitting.

The Watchdog became perfectly still.

For exactly one hundred and eighty-three milliseconds, NatOS did not exist in any meaningful sense.

Then the clock returned.

One tick.

Then another.

The Kernel opened her eyes.

"Status."

The Scheduler answered.

"Clock recovered."

"Memory."

"Intact."

"VM."

"No arena escapes."

"Display."

"Restored."

"Touch."

"Operational."

"Storage."

"Persistent record intact."

"Shell."

"Receiving."

"Watchdog."

"Armed."

The Kernel looked around.

"Cause?"

Nobody answered.

Then the Archivist spoke.

"There is one more thing."

She was standing at the end of the corridor.

The data crystal was still in her hand.

The Kernel walked toward her.

"What?"

The Archivist held it out.

The crystal had changed.

A new record had appeared.

The Kernel read it.

There were only three lines.

BOOT COUNT: 17

FRAME COUNT: 4,812

MESSAGE:

The third line was blank.

The Kernel frowned.

"That's it?"

"No."

The Archivist pointed to the timestamp.

The Kernel read it.

Her expression changed.

The timestamp was impossible.

It was older than the ship.

Older than the operating system.

Older than the hardware.

Older than the civilization that had built NatOS.

The Watchdog stepped forward.

"How old?"

The Archivist answered.

"Four hundred and twelve years."

The Scheduler shook his head.

"Impossible."

The Kernel looked at him.

"That word is banned."

"What?"

"From now on."

She looked at the record again.

"Nothing is impossible until we establish that it cannot happen."

The Watchdog nodded.

The Scheduler frowned.

The VM emerged from his chamber.

The Display illuminated the corridor.

Touch began scanning.

Memory resumed counting.

The Shell began receiving.

The ship slowly came back to life.

Not because the problem was solved.

Because the ship had decided that the problem could be investigated.

The Kernel turned toward the forward observation deck.

The Display showed the stars.

There were thousands.

Millions.

More than anyone aboard could count.

The ship was still inside its construction dock.

It had not yet launched.

It had not yet visited another system.

It had never spoken to another civilization.

It had never fought.

It had never lost anyone.

It had never seen the thing that had written a message into its memory before it was born.

The Kernel stared at the stars.

Then she spoke.

"Prepare for launch."

The Scheduler looked at her.

"We haven't finished verification."

"No."

"We have unresolved hardware questions."

"Yes."

"The persistent record is unexplained."

"Yes."

"The clock anomaly is unexplained."

"Yes."

"The touch calibration is questionable."

"Yes."

"The application containment system has never encountered an actual hostile intelligence."

The Kernel looked at the VM.

He nodded.

"That's correct."

The Kernel turned back toward the stars.

"Then we have everything we need."

The Watchdog smiled.

"What?"

The Kernel's voice was quiet.

"A reason to leave."

Far below them, something inside the ship clicked.

Once.

Then twice.

A signal traveled through the flash controller.

The Archivist looked down.

Her face went pale.

"Kernel."

"What?"

"There is another record."

The Kernel turned.

"Read it."

The Archivist stared at the screen.

"It's not from us."

"Read it."

She swallowed.

Then she spoke.

"Message received."

The entire ship listened.

The message contained only one sentence.

DO NOT LET THE CLOCK STOP.

The lights flickered.

The Scheduler looked at his watch.

For the first time since his awakening, it had hands.

The second hand moved backward.

Then forward.

Then stopped.

The Kernel looked at the Watchdog.

The Watchdog looked at the clock.

Neither spoke.

The ship's engines began to warm.

And somewhere beyond the stars, something noticed that NatOS had finally awakened.